Pre-Calculus 11- Cumulative Exam Part 1
Version 1 - No Calculator

Instructions, read carefully.

1. The formula sheet may be detached but must be handed in with the exam.
2. There are 9 total pages, including this front page and the formula page.
3. Final answers should be exact values or accurate to **two decimal places**.
4. Clearly show your steps or justify your reasoning in the space provided for full marks.
5. Read the box below and then write your name, student number, and signature.

St. George's School - Academic Integrity

Academic Dishonesty includes any conduct with the intent to gain an unfair advantage in connection with an academic assessment. Academic Dishonesty can occur in many ways. Common forms of Academic Dishonesty include, but are not limited to, cheating, falsification, plagiarism, and tampering. Cheating occurs when an individual undermines the integrity of an assessment.

I recognize that the writing of this examination requires an exemplary standard of honesty.

Name : _____ **Student Number :** _____

Signature : _____

Teacher Use only

Total: / 30

Factoring and Radicals

1. [1 point] Which of the following expressions is completely equivalent to $(2x^3)^{-2}$ after applying the exponent laws?

   A. $-4x^6$

   B. $\dfrac{1}{4x^6}$

   C. $\dfrac{1}{2x^6}$

   D. $-2x^{-6}$

2. [1 point] Simplify the mixed radical $\sqrt{72x^3}$, assuming $x \geq 0$.

   A. $36x\sqrt{2x}$

   B. $6x\sqrt{2x}$

   C. $6x^2\sqrt{2}$

   D. $2x\sqrt{6x}$

3. [1 point] Which of the following represents the completely factored form of the difference of squares $4x^2 - 25y^2$?

   A. $(2x - 5y)^2$

   B. $(4x - 5y)(x + 5y)$

   C. $(2x - 5y)(2x + 5y)$

   D. $(2x - 25y)(2x + y)$

4. [2 points] Factor the following trinomial completely.

$$6x^2 - 11x - 10$$

5. [2 points] Simplify the following expression using the exponent laws. Leave your final answer with **positive exponents** only.

$$\frac{(3a^{-2}b^4)^2}{a^3b^{-5}}$$

6. [2 points] Expand and simplify the following radical expression completely.

$$(2\sqrt{3}-\sqrt{5})(3\sqrt{3}+2\sqrt{5})$$

7. [1 point] Rationalize the denominator and simplify the expression. Assume $x > 0$.

$$\frac{6}{\sqrt{3x}}$$

Rational Functions

8. [1 point] What are the non-permissible values (NPVs) for the rational expression $\frac{x+2}{x^2-25}$?

   A. $x \neq -2$
   B. $x \neq 25$
   C. $x \neq 5, x \neq -5$
   D. $x \neq 5$

9. [1 point] Simplify the rational expression $\frac{3x-12}{x-4}$.

   A. $3x$
   B. -3
   C. $x-3$
   D. 3

10. [1 point] What is the lowest common denominator (LCD) needed to add the expressions $\frac{5}{x-2}$ and $\frac{x}{x^2-4}$?

    A. $x-2$
    B. $(x-2)^2$
    C. $(x-2)(x+2)$
    D. $(x-2)(x^2-4)$

11. [2 points] Multiply and simplify the following rational expression completely.

$$\frac{x^2+5x+6}{x^2-4} \cdot \frac{x-2}{3x+9}$$

12. [2 points] Perform the indicated subtraction and simplify.

$$\frac{4}{x+3} - \frac{1}{x-2}$$

13. [2 points] Solve the following rational equation algebraically.

$$\frac{x}{x-3} + \frac{2}{x+3} = \frac{18}{x^2-9}$$

14. [1 point] A boat travels 20 km upstream and 20 km downstream. The speed of the river's current is 2 km/h. Let x represent the speed of the boat in still water. Write a rational expression that represents the **total time** the boat spends traveling. *(Recall that* $Time = \frac{Distance}{Speed}$*).*

Inequalities

15. [1 point] When solving the one-variable linear inequality $-4x + 7 \leq 19$, what is the correct solution?

    A. $x \leq -3$

    B. $x \geq -3$

    C. $x \leq 3$

    D. $x \geq 3$

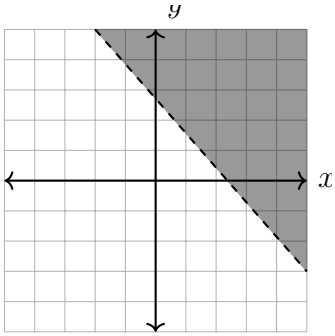
16. [1 point] Which two variable inequality matches the graph below right?

    A. $8x + 7y \leq 19$

    B. $8x + 7y \geq 19$

    C. $8x + 7y > 19$

    D. $8x + 7y < 19$



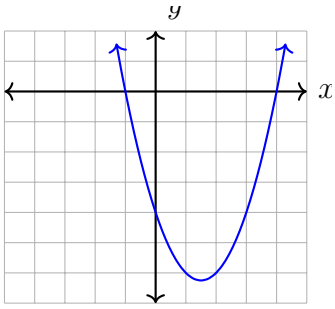
17. [1 point] Using the graph of the quadratic function $f(x) = x^2 - 3x - 4$ below, what is the solution to the one-variable inequality $x^2 - 3x - 10 < 0$?

    A. $x < -1$ or $x > 4$

    B. $-1 < x < 4$

    C. $x \leq -1$ or $x \geq 4$

    D. $-1 \leq x \leq 4$



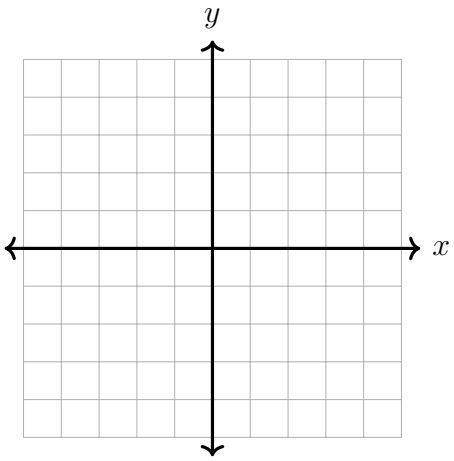
18. [2 points] Solve the following one-variable linear inequality algebraically.

$$-2(x - 4) < 3x + 7$$

19. [2 points] Solve the following one-variable quadratic inequality algebraically.

$$x^2 + x - 12 \geq 0$$

20. [2 points] Sketch the solution region for the two-variable quadratic inequality $y < -x^2+4$ on the grid provided. Be sure to clearly indicate whether the boundary line is solid or dashed.

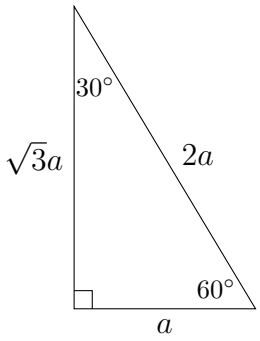


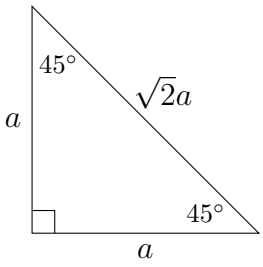
21. [1 point] Determine algebraically whether the point $(3, -2)$ is part of the solution region for the two-variable linear inequality $4x - 3y \geq 18$. Show your test.

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Formula Page

Trigonometry





$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cdot \cos A \qquad \cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

Quadratic Equations

$$a^2 - b^2 = (a + b)(a - b)$$

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Quadratic Functions

$$y = a(x - h)^2 + k$$

$$y = ax^2 + bx + c$$

You may detach this page. Please return it with your exam.